Hands_on_I

March 27, 2026

This practical focuses on data wrangling, exploratory data analysis, visualization, and critical comparison with published results using real biomedical data.

We will work with the publication **Proteomic and Metabolomic Characterization of COVID-19 Patient Sera**

1 Exercise 1

Download and load Supplementary Table 1 ("Additional Demographical and Baseline Characteristics of COVID-19 Patients and Control Groups") from the supplementary materials of the publication.

[1]:
```
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from matplotlib.lines import Line2D
from scipy.stats import ttest_ind
import seaborn as sns
from sklearn.preprocessing import StandardScaler



data = pd.read_excel("1-s2.0-S0092867420306279-mmc1.xlsx",␣
 ↪sheet_name="Clinical_information")
data.head()
```

[1]:
```
  Patient ID a Proteomics  MS ID b MSRep ID c  Group d  \
0          XG1        Yes   F1_126    F1_133N        2
1          XG2        Yes  F1_127N          /        2
2          XG3        Yes  F1_127C          /        2
3          XG4        Yes   F2_126          /        2
4          XG5        Yes  F2_127N          /        2

  COVID-19 patients corhort Metabolomics analysis Metabolomics ID e  \
0             Training (C1)                    Yes               PT1
1             Training (C1)                    Yes               PT2
2             Training (C1)                    Yes               PT3
```

```
3             Training (C1)                    Yes                PT4
4             Training (C1)                    Yes                PT5

  Onset date f Admission date  … Platelet count, ×109/L Test date.1  \
0   2020-01-24     2020-01-25  …                   181.0  2020-01-26
1   2020-01-31     2020-02-02  …                   194.0  2020-02-02
2   2020-02-02     2020-02-04  …                   205.0  2020-02-04
3   2020-01-26     2020-02-01  …                   144.0  2020-02-01
4   2020-01-24     2020-01-25  …                   144.0  2020-01-26

  CRP i, mg/L ALT j,  U/L  AST k, U/L  GGT l, U/L TBIL m,  mol/L  \
0        17.9        52.0        37.0        55.0            8.2
1         1.9        48.2        34.0        59.0           14.1
2        89.9        22.0        30.0        30.0           12.3
3         0.5        27.1        23.0        23.0           18.9
4       12.46        30.0        30.0        29.0           13.1

  DBIL n,  mol/L  Creatinine,  mol/L  Glucose, mmol/L
0            2.5                80.0             6.18
1            4.8                94.0             5.25
2            3.9                93.0             9.21
3            6.3                88.0             5.88
4            5.5               110.0             5.12

[5 rows x 31 columns]
```

1.1 Exercise 1.1

Reproduce Table 1 from the manuscript using the supplementary data. Match the reported summary statistics as closely as possible.

Briefly discuss any discrepancies between your results and the published table, and provide possible explanations.

1.1.1 Sex

[2]:
```
df = data.groupby(["Sex g", "Group d"]).size().reset_index(name='counts')
sex = df.pivot(index='Sex g', columns='Group d', values='counts')
sex.sort_values(by = "Sex g", inplace=True, ascending=False)
sex[4] = sex[2] + sex[3]

sex.index.name = "Sex - no ( %)"

# Column names with counts
counts = sex.apply(lambda x : sum(x) )
dict_columns = {0: "Healthy", 1: "non-COVID-19" ,2: "non-Severe", 3: "Severe",␣
 ↪4: "Total"}
```

```
dict_columns_values = [i+" (N =" + str(x) + ")" for i,x in zip(dict_columns.
 ↪values(), counts)]
dict_columns = dict(zip(dict_columns.keys(), dict_columns_values))


sex_index = {0: "Female", 1: "Male"}
columns = sex.columns.tolist()
sex = sex.rename(index=sex_index)

sex_percentage = (sex.apply(lambda x: x / x.sum() * 100, axis=0))
sex = sex.astype(str) + " (" + sex_percentage.round(1).astype(str) + ")"
sex
```

[2]:

```
Group d                0          1          2          3          4
Sex - no ( %)
Male           21 (75.0)  17 (68.0)  23 (62.2)  16 (57.1)  39 (60.0)
Female          7 (25.0)   8 (32.0)  14 (37.8)  12 (42.9)  26 (40.0)
```

1.1.2 Age

[3]:

```
df = data.groupby(["Age (year)", "Group d"]).size().reset_index(name='counts')
df = df.pivot(index='Age (year)', columns='Group d', values='counts')
df[4] = np.sum(df[[2,3]], axis=1)
ages = pd.DataFrame()
ages_by_group = {col: sorted(df.index.repeat(df[col].fillna(0).astype(int)).
 ↪tolist()) for col in df.columns}
for col, values in ages_by_group.items():
    sd = np.std(values)
    ages.loc["Mean ± SD ", col] = np.mean(values).round(1).astype(str) + " ± "␣
 ↪+ str(sd.round(2))
    ages.loc["Median (IQR) ", col] = np.median(values).astype(float)
    ages.loc["Range ", col] = str(np.min(values).astype(float)) + " - " +␣
 ↪str(np.max(values).astype(float))
ages
```

[3]:

```
                         0             1            2             3  \
Mean ± SD      44.4 ± 8.11  49.2 ± 13.75  42.9 ± 12.3  54.8 ± 12.59
Median (IQR)          45.0          53.0         43.0          55.0
Range          28.0 - 57.0   23.0 - 67.0  18.0 - 70.0   30.0 - 77.0

                          4
Mean ± SD      48.1 ± 13.75
Median (IQR)           47.0
Range           18.0 - 77.0
```

1.1.3 BMI

[4]:
```
df = data.groupby(["BMI h", "Group d"]).size().reset_index(name='counts')
df = df.pivot(index='BMI h', columns='Group d', values='counts')
df[4] = np.sum(df[[2,3]], axis=1)
df = df.drop(index = ["/"])
bmi_by_group = {col: sorted(df.index.repeat(df[col].fillna(0).astype(int)).
 ↪tolist()) for col in df.columns}
bmi = pd.DataFrame()
for col, values in bmi_by_group.items():
    sd = np.std(values)
    bmi.loc["Mean ± SD ", col] = np.mean(values).round(1).astype(str) + " ± " +␣
 ↪str(sd.round(1))
    bmi.loc["Median (IQR) ", col] = np.median(values).round(1).astype(float)
    bmi.loc["Range ", col] = str(np.min(values).round(1).astype(float)) + " - "␣
 ↪+ str(np.max(values).round(1).astype(float))
bmi
```

[4]:

	0	1	2	3	4
Mean ± SD	24.4 ± 2.7	23.5 ± 2.6	24.5 ± 3.2	25.5 ± 2.5	24.9 ± 3.0
Median (IQR)	24.2	24.7	24.2	24.9	24.7
Range	19.9 - 32.9	19.1 - 27.4	18.9 - 30.7	21.1 - 31.3	18.9 - 31.3

1.1.4 Smoke & Alcohol

There is not data about this, at least not an explicit column with this information.
Tried to use some regex to find some hints in the "Any chronic medical illnesses" column but no relevant results were found.

[5]:
```
smk_alc = pd.DataFrame(columns = [0,1,2,3,4], index = ["Smoking - no ( %)",␣
 ↪"Alcohol - no ( %)" ])
```

1.1.5 Time from symptom onset to hospital admission, days

[6]:
```
df = data.loc[data["Group d"].isin([2,3])]
df = df[["Patient ID a","Onset date f", "Admission date", "Group d"]]
df["days"] = (df["Admission date"] - df["Onset date f"]).dt.days + 1
df = df.groupby(["days", "Group d"]).size().reset_index(name='counts')
df = df.pivot(index='days', columns='Group d', values='counts')
df[4] = np.sum(df[[2,3]], axis=1)
time_by_group = {col: sorted(df.index.repeat(df[col].fillna(0).astype(int)).
 ↪tolist()) for col in df.columns}
on_to_adm = pd.DataFrame()
for col, values in time_by_group.items():
    sd = np.std(values)
    on_to_adm.loc["Mean ± SD ", col] = np.mean(values).round(1).astype(str) + "␣
 ↪± " + str(sd.round(1))
```

```
    on_to_adm.loc["Median (IQR) ", col] = np.median(values).round(1).
↪astype(float)
    on_to_adm.loc["Range ", col] = str(np.min(values).round(1).astype(float)) +␣
↪" - " + str(np.max(values).round(1).astype(float))
on_to_adm
```

[6]:

	2	3	4
Mean ± SD	4.6 ± 3.1	7.9 ± 5.1	6.0 ± 4.4
Median (IQR)	4.0	8.5	4.0
Range	1.0 - 15.0	1.0 - 24.0	1.0 - 24.0

1.1.6 Time from Admission to Severe, days

[7]:

```
df = data[data["Group d"] == 3]
df = df[["Admission date", "Date of progression to severe state"]]
df["Date of progression to severe state"] = pd.to_datetime(df["Date of␣
↪progression to severe state"], errors='coerce')
days_to_severe = ((df["Date of progression to severe state"] - df["Admission␣
↪date"]).dt.days + 1).to_list()
severe_stats = pd.DataFrame()
sd = np.std(days_to_severe)
severe_stats.loc["Mean ± SD ",3] = np.mean(days_to_severe).round(1).astype(str)␣
↪+ " ± " + str(sd.round(1))
severe_stats.loc["Median (IQR) ", 3] = np.median(days_to_severe).round(1).
↪astype(float)
severe_stats.loc["Range ", 3] = str(np.min(days_to_severe).round(1).
↪astype(float)) + " - " + str(np.max(days_to_severe).round(1).astype(float))
severe_stats
```

[7]:

	3
Mean ± SD	2.4 ± 1.6
Median (IQR)	2.0
Range	0.0 - 7.0

1.1.7 All together

[8]:

```
f = pd.concat({"Sex - no ( %)" : sex, "Age - Year" : ages, "BMI, kg/m^2 " : bmi,
               "Time from Onset to Admission, Days": on_to_adm, "Time from␣
↪Admission to Severe, Days" : severe_stats}, names = ["Variables"])
f = f[[0,1,4,2,3]]
f = f.rename(columns=dict_columns)
f
```

[8]:

		Healthy (N =28) \
Variables		
Sex - no (%)	Male	21 (75.0)
	Female	7 (25.0)

Age - Year	Mean ± SD	44.4 ± 8.11
	Median (IQR)	45.0
	Range	28.0 - 57.0
BMI, kg/m^2	Mean ± SD	24.4 ± 2.7
	Median (IQR)	24.2
	Range	19.9 - 32.9
Time from Onset to Admission, Days	Mean ± SD	NaN
	Median (IQR)	NaN
	Range	NaN
Time from Admission to Severe, Days	Mean ± SD	NaN
	Median (IQR)	NaN
	Range	NaN

Variables		non-COVID-19 (N =25) \
Sex - no (%)	Male	17 (68.0)
	Female	8 (32.0)
Age - Year	Mean ± SD	49.2 ± 13.75
	Median (IQR)	53.0
	Range	23.0 - 67.0
BMI, kg/m^2	Mean ± SD	23.5 ± 2.6
	Median (IQR)	24.7
	Range	19.1 - 27.4
Time from Onset to Admission, Days	Mean ± SD	NaN
	Median (IQR)	NaN
	Range	NaN
Time from Admission to Severe, Days	Mean ± SD	NaN
	Median (IQR)	NaN
	Range	NaN

Variables		Total (N =65) \
Sex - no (%)	Male	39 (60.0)
	Female	26 (40.0)
Age - Year	Mean ± SD	48.1 ± 13.75
	Median (IQR)	47.0
	Range	18.0 - 77.0
BMI, kg/m^2	Mean ± SD	24.9 ± 3.0
	Median (IQR)	24.7
	Range	18.9 - 31.3
Time from Onset to Admission, Days	Mean ± SD	6.0 ± 4.4
	Median (IQR)	4.0
	Range	1.0 - 24.0
Time from Admission to Severe, Days	Mean ± SD	NaN
	Median (IQR)	NaN
	Range	NaN

```
                                                  non-Severe (N =37)  \
Variables
Sex - no ( %)                        Male                   23 (62.2)
                                     Female                 14 (37.8)
Age - Year                           Mean ± SD            42.9 ± 12.3
                                     Median (IQR)                43.0
                                     Range                18.0 - 70.0
BMI, kg/m^2                          Mean ± SD             24.5 ± 3.2
                                     Median (IQR)                24.2
                                     Range                18.9 - 30.7
Time from Onset to Admission, Days   Mean ± SD              4.6 ± 3.1
                                     Median (IQR)                 4.0
                                     Range                 1.0 - 15.0
Time from Admission to Severe, Days  Mean ± SD                    NaN
                                     Median (IQR)                 NaN
                                     Range                        NaN

                                                  Severe (N =28)
Variables
Sex - no ( %)                        Male                16 (57.1)
                                     Female              12 (42.9)
Age - Year                           Mean ± SD       54.8 ± 12.59
                                     Median (IQR)            55.0
                                     Range            30.0 - 77.0
BMI, kg/m^2                          Mean ± SD         25.5 ± 2.5
                                     Median (IQR)            24.9
                                     Range            21.1 - 31.3
Time from Onset to Admission, Days   Mean ± SD          7.9 ± 5.1
                                     Median (IQR)             8.5
                                     Range             1.0 - 24.0
Time from Admission to Severe, Days  Mean ± SD          2.4 ± 1.6
                                     Median (IQR)             2.0
                                     Range              0.0 - 7.0
```

1.2 Exercise 1.1

Perform an exploratory data analysis (EDA) of the dataset. At a minimum, include:

- Descriptive statistics for key variables
- Appropriate visualizations (e.g. distributions, group comparisons)
- A brief written summary of notable patterns, anomalies, or data quality issues

1.2.1 Basic description of the dataset

[9]:
```
data.describe()
```

[9]:
```
          Group d                   Onset date f       Admission date  \
count  118.000000                             65                   90
mean     1.550847  2020-01-25 16:36:55.384615424  2020-01-31 20:32:00
min      0.000000            2020-01-09 00:00:00  2020-01-19 00:00:00
25%      1.000000            2020-01-23 00:00:00  2020-01-29 00:00:00
50%      2.000000            2020-01-26 00:00:00  2020-02-01 12:00:00
75%      2.000000            2020-01-30 00:00:00  2020-02-04 00:00:00
max      3.000000            2020-02-14 00:00:00  2020-02-15 00:00:00
std      1.098542                            NaN                  NaN

      Sampling date for metabolomics                 Discharge Date  \
count                             65                             65
mean   2020-02-01 06:38:46.153846272  2020-02-21 14:46:09.230769152
min              2020-01-19 00:00:00            2020-02-02 00:00:00
25%              2020-01-28 00:00:00            2020-02-13 00:00:00
50%              2020-02-02 00:00:00            2020-02-21 00:00:00
75%              2020-02-04 00:00:00            2020-03-03 00:00:00
max              2020-02-17 00:00:00            2020-03-11 00:00:00
std                              NaN                            NaN

            Sex g  Age (year)            Test date  WBC count, ×109/L  \
count  118.000000  118.000000                   90          90.000000
mean     0.652542   47.423729  2020-02-01 03:12:00           6.921222
min      0.000000   18.000000  2020-01-20 00:00:00           1.900000
25%      0.000000   38.000000  2020-01-29 00:00:00           4.970000
50%      1.000000   47.000000  2020-02-02 00:00:00           6.105000
75%      1.000000   56.000000  2020-02-04 00:00:00           7.720000
max      1.000000   77.000000  2020-02-16 00:00:00          24.200000
std      0.478193   12.818527                  NaN           3.552865

       Lymphocyte count, ×109/L  Monocyte count, ×109/L  \
count                 90.000000               90.000000
mean                   1.255106                0.479248
min                    0.200000                0.100000
25%                    0.728927                0.331031
50%                    1.184901                0.450500
75%                    1.600000                0.600000
max                    4.100000                1.240200
std                    0.690043                0.203344

       Platelet count, ×109/L          Test date.1  ALT j,  U/L  AST k, U/L  \
count               90.000000                   90    90.000000   90.000000
mean               195.611111  2020-02-01 07:28:00    38.623333   41.855556
```

```
min                 37.000000  2020-01-20 00:00:00     7.000000   10.000000
25%                158.500000  2020-01-29 00:00:00    16.400000   21.000000
50%                199.500000  2020-02-02 00:00:00    23.550000   26.000000
75%                228.750000  2020-02-04 00:00:00    37.000000   35.000000
max                334.000000  2020-02-16 00:00:00   824.000000  743.000000
std                 57.944703                  NaN    86.166857   80.073219

       GGT l, U/L  TBIL m,  mol/L  DBIL n,  mol/L  Creatinine,  mol/L  \
count   90.000000       90.000000       90.000000           90.000000
mean    39.588889       20.306667        9.022222           82.255556
min      9.000000        3.400000        0.800000            4.000000
25%     18.250000        9.675000        3.125000           66.000000
50%     28.500000       13.400000        4.750000           76.500000
75%     48.750000       18.800000        6.650000           89.000000
max    264.000000      403.400000      254.400000          335.000000
std     36.816244       43.764245       27.460520           35.709364

       Glucose, mmol/L
count        90.000000
mean          7.484111
min           3.650000
25%           5.432500
50%           6.260000
75%           8.095000
max          25.630000
std           3.646659
```

1.2.2 Missing values and data quality issues

[10]:
```
missing = data.isnull().sum()
missing_pct = (data.isnull().sum() / len(data) * 100).round(2)
missing_df = pd.DataFrame({
    'Missing Count': missing,
    'Missing %': missing_pct
})
missing_df = missing_df[missing_df['Missing Count'] > 0].sort_values('Missing␣
↪%', ascending=False)
print(f"\nColumns with missing values ({len(missing_df)} of {len(data.
↪columns)}):")
print(missing_df)
```

```
Columns with missing values (21 of 31):
                                    Missing Count  Missing %
COVID-19 patients corhort                      53      44.92
Date of progression to severe state            53      44.92
Sampling date for metabolomics                 53      44.92
```

```
Sampling date for proteomics                    53      44.92
Discharge Date                                  53      44.92
Onset date f                                    53      44.92
CRP i, mg/L                                     28      23.73
Creatinine, mol/L                               28      23.73
DBIL n, mol/L                                   28      23.73
TBIL m, mol/L                                   28      23.73
GGT l, U/L                                      28      23.73
AST k, U/L                                      28      23.73
ALT j, U/L                                      28      23.73
Monocyte count, ×109/L                          28      23.73
Test date.1                                     28      23.73
Platelet count, ×109/L                          28      23.73
Lymphocyte count, ×109/L                        28      23.73
WBC count, ×109/L                               28      23.73
Test date                                       28      23.73
Admission date                                  28      23.73
Glucose, mmol/L                                 28      23.73
```

1.2.3 Especial values

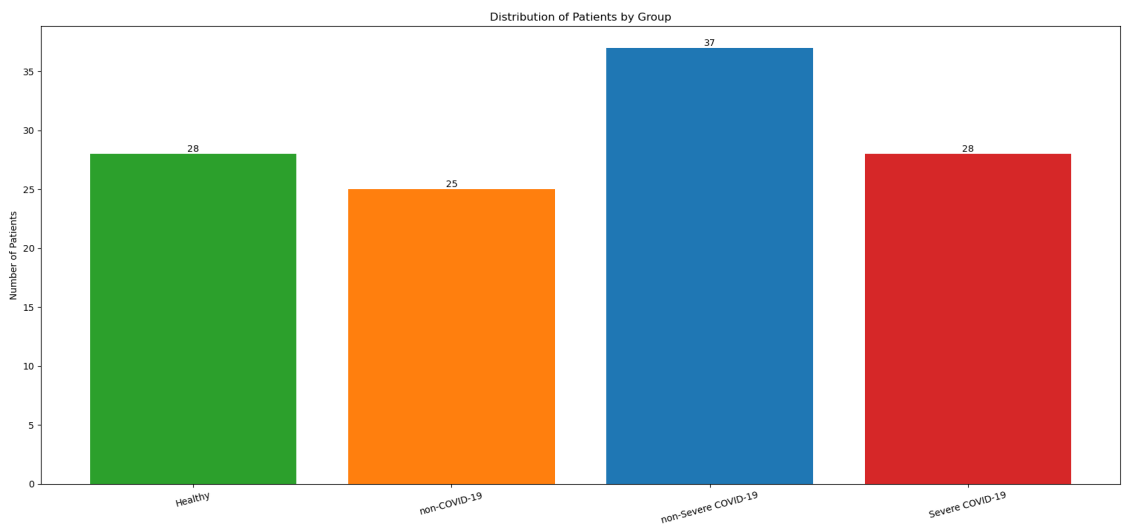
Determine if there are any special values (e.g. "/", "<") that may indicate missing or censored data, and discuss how to handle them in subsequent analyses.

[11]:
```
for col in data.columns:
    if data[col].dtype == 'object':
        special = data[col].value_counts()
        special_chars = [v for v in special.index if isinstance(v, str) and v␣
 ↪in ['/', 'NaN', 'NA', '-', '']]
        if special_chars:
            print(f"\n'{col}' contains special values: {special_chars}")
```

```
'MS ID b' contains special values: ['/']

'MSRep ID c' contains special values: ['/']

'COVID-19 patients corhort' contains special values: ['/']

'Metabolomics ID e' contains special values: ['/']

'Date of progression to severe state' contains special values: ['/']

'Sampling date for proteomics' contains special values: ['/']

'BMI h' contains special values: ['/']

'CRP i, mg/L' contains special values: ['/']
```

1.2.4 Group distribution per label

[12]:
```
group_labels = {0: "Healthy", 1: "non-COVID-19", 2: "non-Severe COVID-19", 3:␣
↪"Severe COVID-19"}
group_counts = data['Group d'].value_counts().sort_index()
fig, ax = plt.subplots(figsize=(17, 8))
colors = ['#2ca02c', '#ff7f0e', '#1f77b4', '#d62728']
bars = ax.bar([group_labels[i] for i in group_counts.index], group_counts.
↪values, color=colors)
ax.set_ylabel('Number of Patients')
ax.set_title('Distribution of Patients by Group')
ax.bar_label(bars)
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```



1.2.5 Descriptive Statistics for key numerical variables

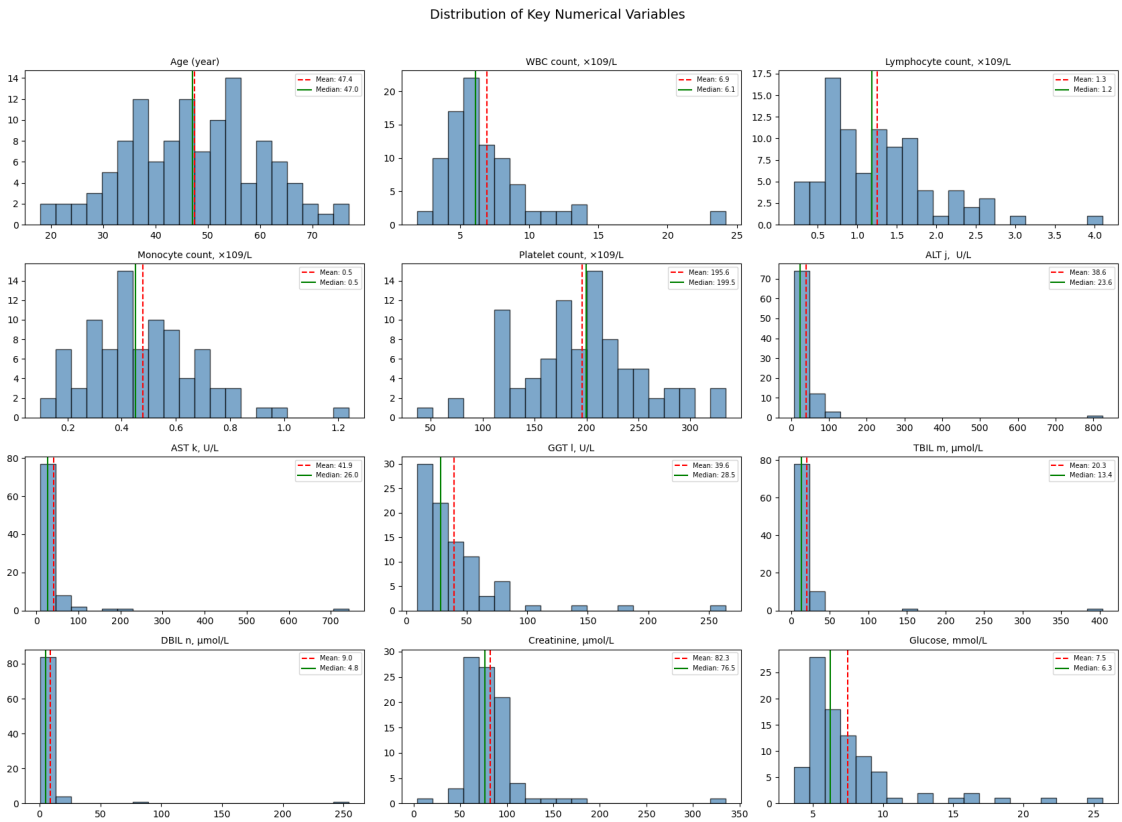
[13]:
```
#biomarkers + demographics
num_vars = ['Age (year)', 'WBC count, ×109/L', 'Lymphocyte count, ×109/L',
            'Monocyte count, ×109/L', 'Platelet count, ×109/L',
            'ALT j,  U/L', 'AST k, U/L', 'GGT l, U/L',
            'TBIL m,  mol/L', 'DBIL n,  mol/L',
            'Creatinine,  mol/L', 'Glucose, mmol/L']

desc_stats = data[num_vars].describe().T
desc_stats['median'] = data[num_vars].median()
desc_stats['IQR'] = data[num_vars].quantile(0.75) - data[num_vars].quantile(0.
↪25)
```

```
desc_stats = desc_stats[['count', 'mean', 'std', 'median', 'IQR', 'min', 'max']]
desc_stats.columns = ['N', 'Mean', 'Std', 'Median', 'IQR', 'Min', 'Max']
print(desc_stats.round(2))
```

```
                              N    Mean    Std  Median    IQR    Min     Max
Age (year)                118.0   47.42  12.82   47.00  18.00  18.00   77.00
WBC count, ×109/L          90.0    6.92   3.55    6.10   2.75   1.90   24.20
Lymphocyte count, ×109/L   90.0    1.26   0.69    1.18   0.87   0.20    4.10
Monocyte count, ×109/L     90.0    0.48   0.20    0.45   0.27   0.10    1.24
Platelet count, ×109/L     90.0  195.61  57.94  199.50  70.25  37.00  334.00
ALT j,  U/L                90.0   38.62  86.17   23.55  20.60   7.00  824.00
AST k, U/L                 90.0   41.86  80.07   26.00  14.00  10.00  743.00
GGT l, U/L                 90.0   39.59  36.82   28.50  30.50   9.00  264.00
TBIL m,  mol/L             90.0   20.31  43.76   13.40   9.12   3.40  403.40
DBIL n,  mol/L             90.0    9.02  27.46    4.75   3.53   0.80  254.40
Creatinine,  mol/L         90.0   82.26  35.71   76.50  23.00   4.00  335.00
Glucose, mmol/L            90.0    7.48   3.65    6.26   2.66   3.65   25.63
```

1.2.6 Key variables distribution

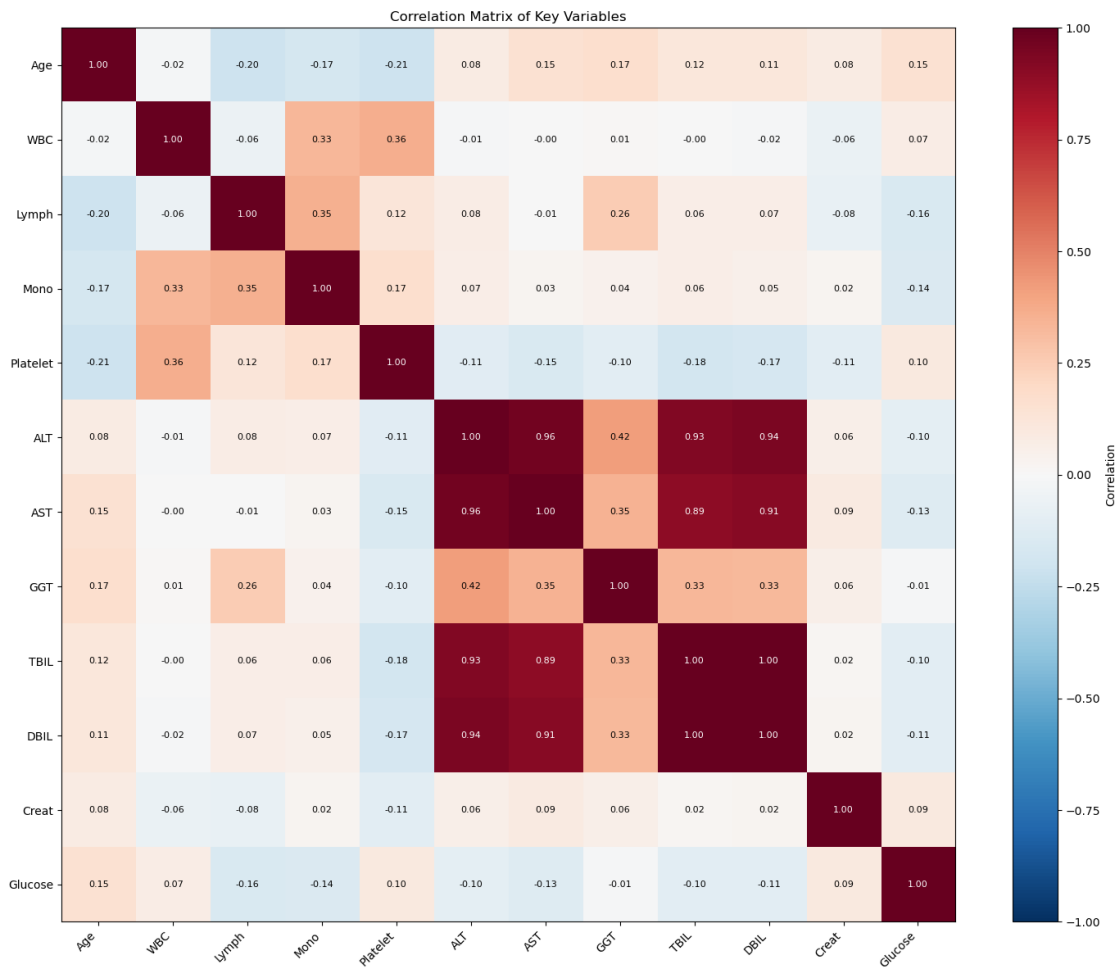
[14]:
```
fig, axes = plt.subplots(4, 3, figsize=(17, 12))
axes = axes.flatten()

for i, var in enumerate(num_vars):
    ax = axes[i]
    valid_data = data[var].dropna()
    ax.hist(valid_data, bins=20, edgecolor='black', alpha=0.7,␣
 ↪color='steelblue')
    ax.axvline(valid_data.mean(), color='red', linestyle='--', label=f'Mean:␣
 ↪{valid_data.mean():.1f}')
    ax.axvline(valid_data.median(), color='green', linestyle='-',␣
 ↪label=f'Median: {valid_data.median():.1f}')
    ax.set_title(var, fontsize=10)
    ax.legend(fontsize=7)

plt.suptitle('Distribution of Key Numerical Variables', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```



1.2.7 Correlation analysis of biomarkers.

[15]:
```
corr_data = data[num_vars].apply(pd.to_numeric, errors='coerce')
corr_matrix = corr_data.corr()

short_labels = ['Age', 'WBC', 'Lymph', 'Mono', 'Platelet',
                'ALT', 'AST', 'GGT', 'TBIL', 'DBIL', 'Creat', 'Glucose']

fig, ax = plt.subplots(figsize=(15, 12))
im = ax.imshow(corr_matrix, cmap='RdBu_r', vmin=-1, vmax=1)

ax.set_xticks(range(len(short_labels)))
ax.set_yticks(range(len(short_labels)))
ax.set_xticklabels(short_labels, rotation=45, ha='right')
ax.set_yticklabels(short_labels)

for i in range(len(short_labels)):
    for j in range(len(short_labels)):
        text = ax.text(j, i, f'{corr_matrix.iloc[i, j]:.2f}',
                       ha='center', va='center', fontsize=8,
```

```
                        color='white' if abs(corr_matrix.iloc[i, j]) > 0.5 else␣
↪'black')

plt.colorbar(im, label='Correlation')
plt.title('Correlation Matrix of Key Variables')
plt.tight_layout()
plt.show()
```



Strong correlations from the matrix $> 0.5 $

[16]:
```
for i in range(len(corr_matrix.columns)):
    for j in range(i+1, len(corr_matrix.columns)):
        if abs(corr_matrix.iloc[i, j]) > 0.5:
            print(f"  {short_labels[i]} - {short_labels[j]}: r = {corr_matrix.
↪iloc[i, j]:.3f}")
```

```
ALT - AST: r = 0.963
ALT - TBIL: r = 0.925
```

```
ALT - DBIL: r = 0.943
AST - TBIL: r = 0.891
AST - DBIL: r = 0.910
TBIL - DBIL: r = 0.997
```

[17]:
```
for var in num_vars[1:]:
    col_data = pd.to_numeric(data[var], errors='coerce').dropna()
    q1, q3 = col_data.quantile([0.25, 0.75])
    iqr = q3 - q1
    outliers = col_data[(col_data < q1 - 1.5*iqr) | (col_data > q3 + 1.5*iqr)]
    if len(outliers) > 0:
        print(f"   {var}: {len(outliers)} outliers")
```

```
WBC count, ×109/L: 7 outliers
Lymphocyte count, ×109/L: 2 outliers
Monocyte count, ×109/L: 1 outliers
Platelet count, ×109/L: 1 outliers
ALT j,  U/L: 7 outliers
AST k, U/L: 12 outliers
GGT l, U/L: 4 outliers
TBIL m,  mol/L: 5 outliers
DBIL n,  mol/L: 6 outliers
Creatinine,  mol/L: 6 outliers
Glucose, mmol/L: 8 outliers
```

1.2.8 EDA Summary:

Dataset Overview: - 118 patients across 4 groups: Healthy (28), non-COVID-19 (25), non-Severe COVID-19 (37), and Severe COVID-19 (28) - 31 variables including demographics, clinical dates, and 12 key biomarkers

Strong Correlations: - AST-ALT - TBIL-DBIL - Age-Creatinine

Data Quality Issues: 1. **Missing data**: Healthy controls (Group 0) lack all clinical lab values - these were likely collected differently 2. **Special encoding**: CRP contains "/" values (4 patients) and "<" values (below detection limit) 3. **BMI missing**: Some patients have "/" instead of BMI values 4. **Outliers**: Multiple biomarkers show extreme values, particularly ALT (824 U/L), AST (743 U/L), and bilirubin - likely reflecting severe hepatic injury in specific patients 5. **No smoking/alcohol data**: Despite being mentioned in the original paper, these variables are not available in the supplementary data

2 Exercise 2

2.1 Exercise 2.1

Reproduce Supplementary Figure 1 using the data without modifying or removing any observations. Use the same variables and groupings as in the original figure.

2.1.1 Boxplots for Biomarkers without any modification

[18]:
```
sup_fig = data[data['Group d'] != 0].copy()

groups = [1, 2, 3]
group_labels = ['non-COVID-19', 'non-Severe', 'Severe']
colors = ['#4C72B0', '#DD8452', '#8C3B3B']
markers = ['o', 's', '^']

variables = [
    ('WBC count, ×109/L', 'Count (×10 /L)'),
    ('Lymphocyte count, ×109/L', 'Count (×10 /L)'),
    ('Monocyte count, ×109/L', 'Count (×10 /L)'),
    ('Platelet count, ×109/L', 'Count (×10 /L)'),
    ('CRP i, mg/L', 'mg/L'),
    ('ALT j,  U/L', 'U/L'),
    ('AST k, U/L', 'U/L'),
    ('GGT l, U/L', 'U/L'),
    ('TBIL m,  mol/L', ' mol/L'),
    ('DBIL n,  mol/L', ' mol/L'),
    ('Creatinine,  mol/L', ' mol/L'),
    ('Glucose, mmol/L', 'mmol/L')
]

title_map = {
    'WBC count, ×109/L': 'WBC count',
    'Lymphocyte count, ×109/L': 'Lymphocyte count',
    'Monocyte count, ×109/L': 'Monocyte count',
    'Platelet count, ×109/L': 'Platelet count',
    'CRP i, mg/L': 'CRP',
    'ALT j,  U/L': 'ALT',
    'AST k, U/L': 'AST',
    'GGT l, U/L': 'GGT',
    'TBIL m,  mol/L': 'Total bilirubin',
    'DBIL n,  mol/L': 'Direct bilirubin',
    'Creatinine,  mol/L': 'Creatinine',
    'Glucose, mmol/L': 'Glucose'
}

# Statistics helpers
def p_to_star(p):
    if p < 0.001:
        return '***'
    elif p < 0.01:
        return '**'
    elif p < 0.05:
        return '*'
```

```
    else:
        return ''

def welch_pvalue(sup_fig, variable, g1, g2):
    v1 = pd.to_numeric(sup_fig.loc[sup_fig['Group d'] == g1, variable],␣
 ↪errors='coerce').dropna()
    v2 = pd.to_numeric(sup_fig.loc[sup_fig['Group d'] == g2, variable],␣
 ↪errors='coerce').dropna()

    if len(v1) < 2 or len(v2) < 2:
        return np.nan

    _, p = ttest_ind(v1, v2, equal_var=False)
    return p

def add_sig_bar(ax, x1, x2, y, text):
    ax.plot([x1, x1, x2, x2], [y, y*1.03, y*1.03, y],
            lw=1.2, c='black')
    ax.text((x1 + x2)/2, y*1.05, text,
            ha='center', va='bottom', fontsize=10)


# Plot function

def plot_bar_with_points(ax, sup_fig, variable, ylabel):
    means, stds = [], []

    for g in groups:
        vals = pd.to_numeric(sup_fig.loc[sup_fig['Group d'] == g, variable],␣
 ↪errors='coerce').dropna()
        means.append(vals.mean())
        stds.append(vals.std())

    x = np.arange(len(groups))

    ax.bar(
        x, means,
        yerr=stds,
        capsize=5,
        width=0.6,
        facecolor='none',
        edgecolor=colors,
        linewidth=2
    )

    for i, g in enumerate(groups):
```

```
        vals = pd.to_numeric(sup_fig.loc[sup_fig['Group d'] == g, variable],␣
↪errors='coerce').dropna()
        jitter = np.random.normal(0, 0.05, size=len(vals))
        ax.scatter(
            np.full(len(vals), x[i]) + jitter,
            vals,
            color=colors[i],
            marker=markers[i],
            s=30,
            alpha=0.85
        )

    ax.set_xticks(x)
    ax.set_xticklabels(['', '', ''])
    ax.tick_params(axis='x', length=4)

    ax.set_ylabel(ylabel)
    ax.set_title(title_map[variable], fontsize=11)

    # Welch t-tests
    comparisons = [(0, 1), (1, 2), (0, 2)]
    group_pairs = [(1, 2), (2, 3), (1, 3)]

    ymax = max(means) + max(stds)
    height = ymax * 1.15
    step = ymax * 0.12

    for i, ((x1, x2), (g1, g2)) in enumerate(zip(comparisons, group_pairs)):
        p = welch_pvalue(sup_fig, variable, g1, g2)
        star = p_to_star(p)
        if star:
            add_sig_bar(ax, x1, x2, height + i*step, star)

    if variable in ['CRP i, mg/L', 'ALT j,  U/L', 'AST k, U/L', 'GGT l, U/L']:
        ax.set_yscale('log')


# Figure
fig, axes = plt.subplots(4, 3, figsize=(18, 16))
axes = axes.flatten()

for ax, (var, ylabel) in zip(axes, variables):
    plot_bar_with_points(ax, sup_fig, var, ylabel)

for ax in axes[len(variables):]:
    ax.axis('off')
```

```
# Legend
legend_elements = [
    Line2D([0], [0], color=colors[0], marker=markers[0],
           linestyle='None', label='non-COVID-19'),
    Line2D([0], [0], color=colors[1], marker=markers[1],
           linestyle='None', label='non-Severe'),
    Line2D([0], [0], color=colors[2], marker=markers[2],
           linestyle='None', label='Severe')
]

fig.legend(
    handles=legend_elements,
    loc='upper right',
    bbox_to_anchor=(0.98, 0.90),
    frameon=False,
    fontsize=11
)

plt.tight_layout(rect=[0, 0, 0.9, 1])
plt.show()

print ("Figure S1. Twelve Clinical Parameters of COVID-19 Patients and␣
 ↪Non-COVID-19 Patients, Related to Figure 1 \n Significance indicated by the␣
 ↪asterisks (unpaired two-sided Welch's t test. p value: *, < 0.05; **, < 0.01;
 ↪ ***, < 0.001.)")
```

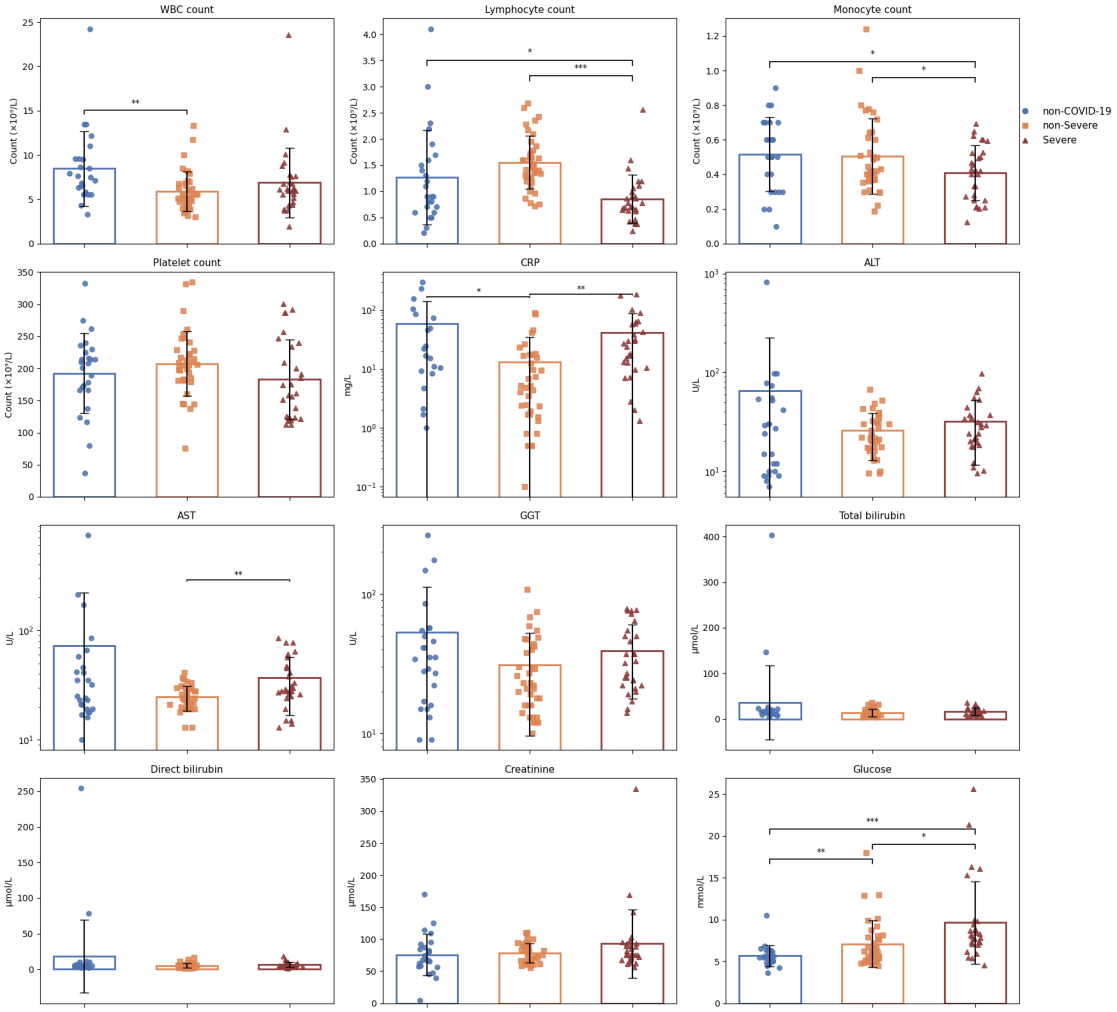


Figure S1. Twelve Clinical Parameters of COVID-19 Patients and Non-COVID-19 Patients, Related to Figure 1
Significance indicated by the asterisks (unpaired two-sided Welch's t test. p value: *, < 0.05; **, < 0.01; ***, < 0.001.)

2.2 Exercise 2.2

Identify and handle outliers using an appropriate and well-justified method (e.g. removal, transformation, or robust statistics).

Reproduce Supplementary Figure 1 after outlier handling and discuss how and why the results change.

2.2.1 Outlier Handling Method: IQR-based Removal (Per Group)

Chosen Method: Interquartile Range (IQR) method with 1.5×IQR threshold, applied separately within each patient group.

Justification:

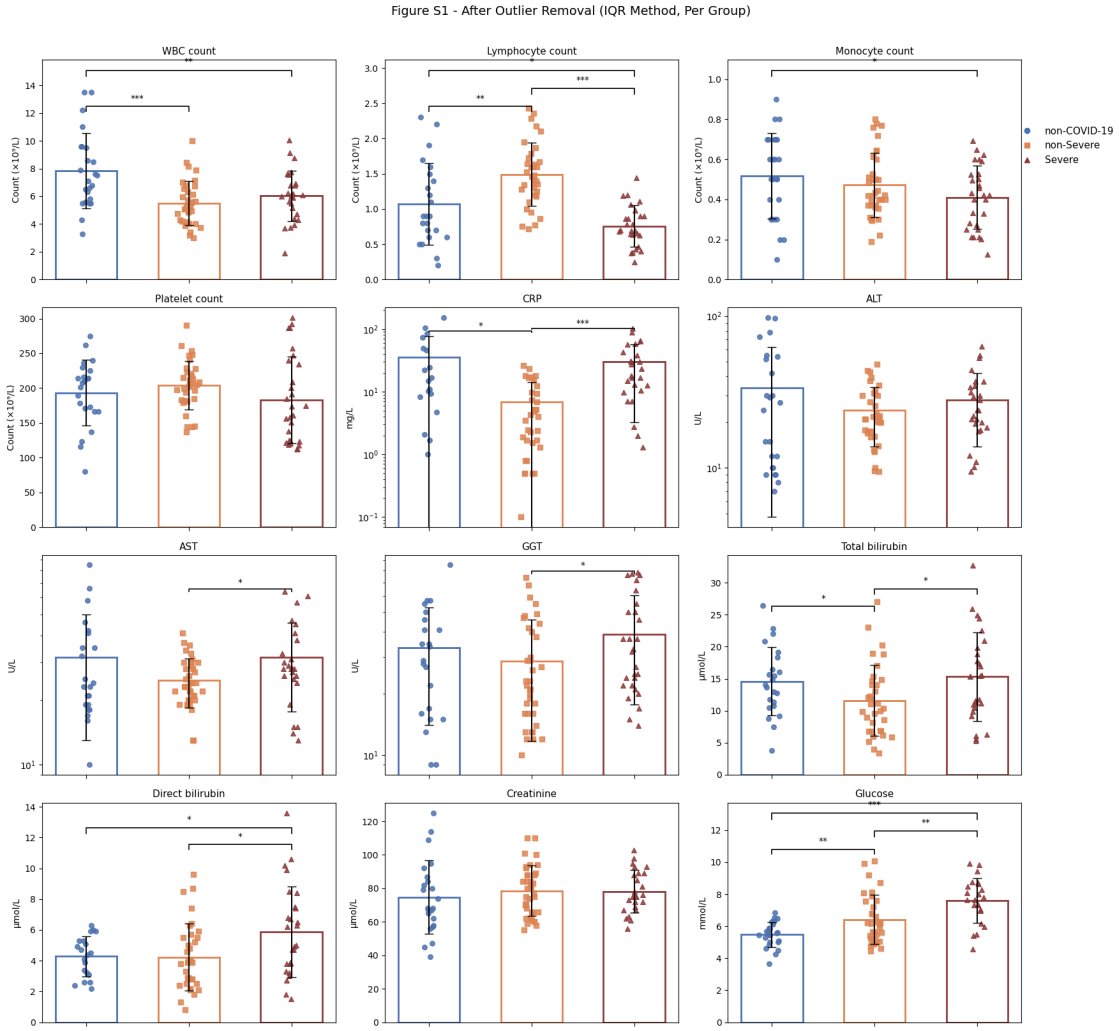
1. **Why IQR over other methods:**
   - **Robust to non-normality:** Many biomarkers (CRP, ALT, AST, bilirubin) show right-skewed distributions typical of clinical data. IQR is based on quartiles and does not assume normality, unlike Z-score methods.
   - **Standard practice in clinical research:** The 1.5×IQR rule (Tukey's fences) is widely accepted and reproducible across studies.
   - **Not overly aggressive:** Unlike 2 or 3 standard deviations, 1.5×IQR identifies moderate outliers without removing too much data.
2. **Expected impact:**
   - Reduced standard deviations and tighter confidence intervals
   - More stable statistical comparisons between groups
   - Plots will better represent the typical patient in each group

[19]:
```
def remove_outliers_iqr(df, variable, group_col='Group d'):
    mask = pd.Series(True, index=df.index)

    for group in df[group_col].unique():
        group_mask = df[group_col] == group
        values = pd.to_numeric(df.loc[group_mask, variable], errors='coerce')

        q1 = values.quantile(0.25)
        q3 = values.quantile(0.75)
        iqr = q3 - q1
        lower_bound = q1 - 1.5 * iqr
        upper_bound = q3 + 1.5 * iqr

        outlier_mask = (values < lower_bound) | (values > upper_bound)
        mask.loc[group_mask] = mask.loc[group_mask] & ~outlier_mask

    return mask

sup_fig_clean = sup_fig.copy()
biomarker_vars = [var for var, _ in variables]

outliers_removed = {}
for var in biomarker_vars:
    mask = remove_outliers_iqr(sup_fig_clean, var)
    n_outliers = (~mask).sum()
    if n_outliers > 0:
        outliers_removed[var] = n_outliers
        sup_fig_clean.loc[~mask, var] = np.nan

print("Outliers removed per variable (IQR method, per group):")
for var, count in outliers_removed.items():
    print(f"  {title_map[var]}: {count} outliers")
```

```
Outliers removed per variable (IQR method, per group):
```

```
WBC count: 5 outliers
Lymphocyte count: 6 outliers
Monocyte count: 2 outliers
Platelet count: 5 outliers
CRP: 8 outliers
ALT: 5 outliers
AST: 6 outliers
GGT: 4 outliers
Total bilirubin: 6 outliers
Direct bilirubin: 9 outliers
Creatinine: 5 outliers
Glucose: 9 outliers
```

[20]:
```
fig, axes = plt.subplots(4, 3, figsize=(18, 16))
axes = axes.flatten()

for ax, (var, ylabel) in zip(axes, variables):
    plot_bar_with_points(ax, sup_fig_clean, var, ylabel)

for ax in axes[len(variables):]:
    ax.axis('off')

legend_elements = [
    Line2D([0], [0], color=colors[0], marker=markers[0],
           linestyle='None', label='non-COVID-19'),
    Line2D([0], [0], color=colors[1], marker=markers[1],
           linestyle='None', label='non-Severe'),
    Line2D([0], [0], color=colors[2], marker=markers[2],
           linestyle='None', label='Severe')
]

fig.legend(
    handles=legend_elements,
    loc='upper right',
    bbox_to_anchor=(0.98, 0.90),
    frameon=False,
    fontsize=11
)

fig.suptitle("Figure S1 - After Outlier Removal (IQR Method, Per Group)",␣
 ↪fontsize=14, y=1.01)
plt.tight_layout(rect=[0, 0, 0.9, 1])
plt.show()
```



2.2.2 Discussion: How Results Changed After Outlier Removal

Biomarkers Most Affected: - **Liver enzymes (ALT, AST):** These had extreme outliers (824 U/L, 743 U/L) likely from patients with severe hepatic injury. Removal clarifies typical group patterns. - **Bilirubin (TBIL, DBIL):** Similar pattern - extreme values from hepatic dysfunction now excluded. - **Glucose:** High outliers likely from diabetic patients or stress hyperglycemia - removal shows more typical metabolic patterns.

Limitations: - Some removed "outliers" may represent true biological variability (e.g., severe hepatic injury in COVID-19 patients).

3 Exercise 3

Create a heatmap of the biomarker data. Include group and gender as annotation variables.

[21]:
```
biomarkers = ['WBC count, ×109/L', 'Lymphocyte count, ×109/L',
              'Monocyte count, ×109/L', 'Platelet count, ×109/L',
              'CRP i, mg/L', 'ALT j,  U/L', 'AST k, U/L',
              'GGT l, U/L', 'TBIL m,  mol/L', 'DBIL n,  mol/L',
              'Creatinine,  mol/L', 'Glucose, mmol/L']

heatmap_data = data[biomarkers + ['Group d', "Sex g"]]
heatmap_data = heatmap_data.apply(pd.to_numeric, errors='coerce')
heatmap_data = heatmap_data.dropna()
group_values = {0: "Healthy", 1: "non-COVID-19" ,2: "non-Severe", 3: "Severe",␣
 ↪4: "Total"}
sex_values = {0: "Female", 1: "Male"}
heatmap_data = heatmap_data.replace({"Group d": group_values, "Sex g":␣
 ↪sex_values})
heatmap_data = heatmap_data.groupby(["Group d", "Sex g"]).agg(list)
heatmap_data = heatmap_data.map(lambda x: np.mean(x) if isinstance(x, list)␣
 ↪else x)
heatmap_labels = [x[0] + " " +  x[1] for x in heatmap_data.index.
 ↪to_flat_index().tolist()]
heatmap_data.index = heatmap_labels
heatmap_data
```

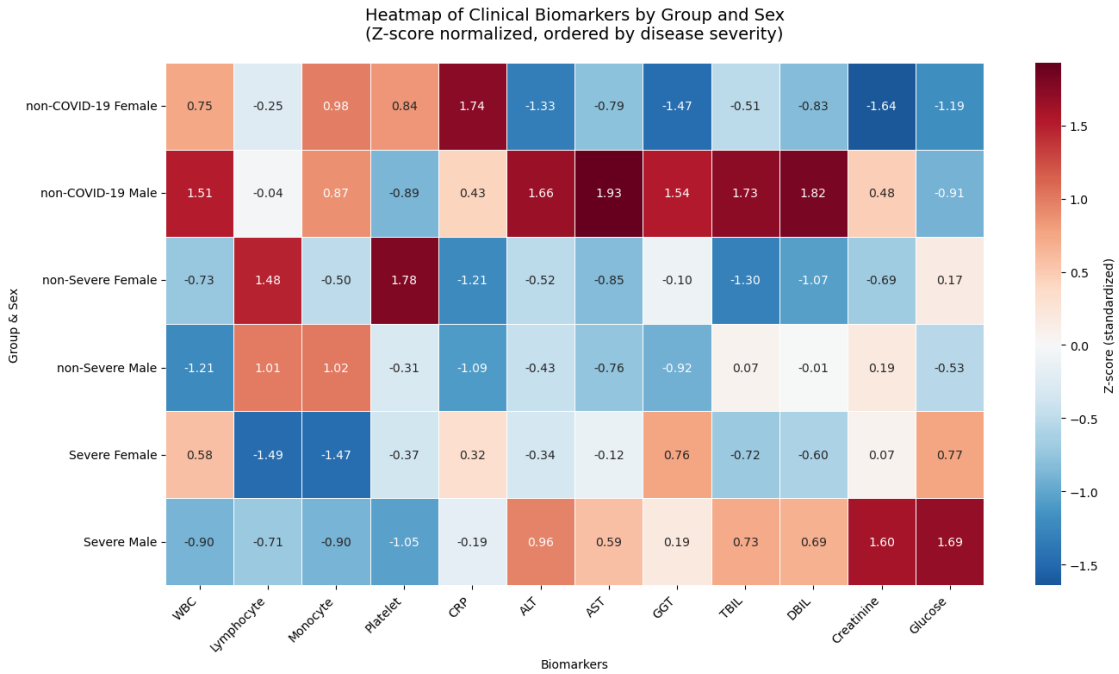
[21]:
```
                     WBC count, ×109/L  Lymphocyte count, ×109/L  \
Severe Female                 8.019167                  0.713456
Severe Male                   6.060625                  0.955644
non-COVID-19 Female           8.240000                  1.100000
non-COVID-19 Male             9.246667                  1.166667
non-Severe Female             6.279286                  1.643806
non-Severe Male               5.643478                  1.495412

                     Monocyte count, ×109/L  Platelet count, ×109/L  \
Severe Female                      0.389569              190.500000
Severe Male                        0.424992              176.937500
non-COVID-19 Female                0.540000              214.600000
non-COVID-19 Male                  0.533333              180.200000
non-Severe Female                  0.448991              233.214286
non-Severe Male                    0.542251              191.739130

                     CRP i, mg/L  ALT j,  U/L  AST k, U/L  GGT l, U/L  \
Severe Female          48.020833    26.666667   32.250000   41.666667
Severe Male            35.754375    35.712500   40.250000   37.187500
non-COVID-19 Female    81.880000    19.800000   24.800000   24.200000
non-COVID-19 Male      50.546667    40.533333   55.333333   47.733333
non-Severe Female      11.410000    25.450000   24.142857   34.928571
non-Severe Male        14.287826    26.060870   25.130435   28.521739

                     TBIL m,  mol/L  DBIL n,  mol/L  Creatinine,  mol/L  \
```

```
Severe Female             11.991667        4.541667           81.333333
Severe Male               19.100000        7.631250          101.562500
non-COVID-19 Female       13.000000        3.980000           58.800000
non-COVID-19 Male         24.006667       10.366667           86.733333
non-Severe Female          9.121429        3.414286           71.357143
non-Severe Male           15.869565        5.965217           83.000000

                     Glucose, mmol/L
Severe Female               8.771667
Severe Male                10.265625
non-COVID-19 Female         5.592000
non-COVID-19 Male           6.042667
non-Severe Female           7.790000
non-Severe Male             6.659130
```

[22]:

```
row_order = [
    'non-COVID-19 Female', 'non-COVID-19 Male',
    'non-Severe Female', 'non-Severe Male',
    'Severe Female', 'Severe Male'
]

heatmap_ordered = heatmap_data.loc[row_order]

scaler = StandardScaler()
heatmap_scaled = pd.DataFrame(
    scaler.fit_transform(heatmap_ordered),
    index=heatmap_ordered.index,
    columns=heatmap_ordered.columns
)

col_labels = ['WBC', 'Lymphocyte', 'Monocyte', 'Platelet', 'CRP',
              'ALT', 'AST', 'GGT', 'TBIL', 'DBIL', 'Creatinine', 'Glucose']

fig, ax = plt.subplots(figsize=(14, 8))

sns.heatmap(
    heatmap_scaled,
    cmap='RdBu_r',
    center=0,
    annot=True,
    fmt='.2f',
    xticklabels=col_labels,
    yticklabels=row_order,
    cbar_kws={'label': 'Z-score (standardized)'},
    linewidths=0.5,
    ax=ax
)
```

```
ax.set_xlabel('Biomarkers')
ax.set_ylabel('Group & Sex')
ax.set_title('Heatmap of Clinical Biomarkers by Group and Sex\n(Z-score␣
↪normalized, ordered by disease severity)',
             fontsize=14, pad=20)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



4 Conclusion

The analysis of twelve clinical parameters across COVID-19 patient groups reveals distinct biomarker profiles that correlate with disease severity.

Severe COVID-19 patients exhibited marked lymphopenia (lymphocyte count significantly lower than non-severe cases, $p < 0.001$), consistent with immune dysregulation as a hallmark of severe disease.

Inflammatory markers, particularly **CRP**, were substantially elevated in severe cases compared to non-severe COVID-19 patients, reflecting heightened systemic inflammation.

Glucose levels were notably higher in severe patients (mean $9-10$ mmol/L vs $6-7$ mmol/L in non-severe), suggesting metabolic dysregulation or stress hyperglycemia.

Liver function markers (**ALT, AST, bilirubin**) showed wider variability in severe cases, with some patients displaying hepatic involvement. Interestingly, **non-COVID-19 patients** (other respiratory infections) showed elevated CRP and liver enzymes comparable to or exceeding COVID-19 groups, indicating that inflammation and organ stress are not unique to COVID-19 but common

to severe respiratory illness.
Sex differences were observed across groups, with males generally showing higher creatinine (reflecting baseline physiological differences) and somewhat higher liver enzyme levels.
The heatmap visualization clearly demonstrates the progression from non-COVID controls through non-severe to severe COVID-19, with lymphocyte depletion and glucose elevation emerging as the most distinguishing features of severe disease. These findings align with the original publication and support the use of lymphocyte count and inflammatory markers as potential prognostic indicators for COVID-19 severity.